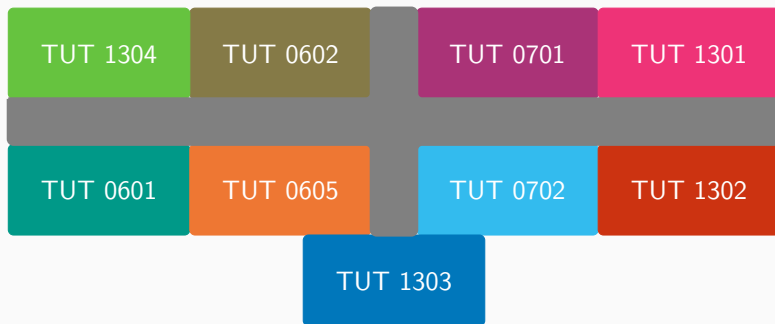


Front Screens



Lecture 6 Instantaneous Rate of Change

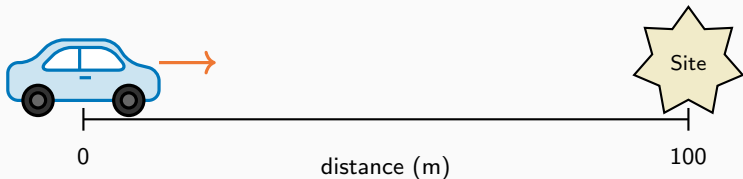
Big Question

- How can we describe a rate of change in a single instant?

Warm-up: Safe Road Design

Driving speed is an important factor in road safety. It affects the severity of a crash and is also related to the risk of being involved in a crash.

You've been tasked to study a particular common crash site on a road to find how speed relates to risk of crashes. You have video footage showing cars driving through the crash site and the 100 metres leading up to it.



Discussion Question

Using nothing but this video footage, how might you estimate the speed of each car at the crash site? (Assume you have no odometers, sensors, or AI to help you!)

One Possible method: In the video, label the location that's 100 metres before the crash site. Measure the time it takes for the car to travel those 100 metres. Use this information.

Poll Question

If the car takes 5 seconds to travel these 100 metres, what can you say about the car's speed *exactly* as it passes through the crash site? (This car doesn't crash.)

- ☒ **A** $> 20 \text{ m/s}$
- ☐ **B** $< 20 \text{ m/s}$
- ☐ **C** $= 20 \text{ m/s}$
- ☐ **D** I don't know.

Discussion Question

Can you suggest an improvement on the method described?

Improve the precision: Let's use the video to collect data at more times, especially closer to time $t = 5$ seconds.

Time t (s)	0	4	4.9	4.99	4.999	5	5.001	5.01	5.1	6
Distance x (m)	0	64	96.04	99.6	99.95	100	100.04	100.4	104.4	144

Discussion Question

How can we use this data to improve our estimate for the speed of the car at time $t = 5$ seconds? Come up with at least two methods. Using a method discussed by your group, find an estimate.

Instantaneous velocity

We can use our idea from the previous slide to *define* instantaneous velocity.

Definition

The **instantaneous velocity** of an object at time t is defined to be the **limit** of the of the object over shorter and shorter time intervals containing t .

Instantaneous velocity using the graph

Let's suppose that we can also use the video to plot a graph of the car's position as a function of time, shown in the Desmos graph linked below.

<https://www.desmos.com/calculator/e84ktt0goy>

Discussion Question

- (a) Adjust the positions of the two points so that the slope of the secant line represents the average velocity over the first 5s. (The average velocity is m in the left panel.)
- (b) Make further adjustments to the graph to more accurately estimate the velocity of the car exactly at $t = 5$ s.

Why do we care about velocity?

Why velocity?

- We will often seek to apply calculus to commerce and social sciences in this course!
- However, physical motion is easier for us to conceptualize, as humans.
- We'll often use physical metaphors as we think about other applications.

Instantaneous Rates of Change

More generally, we can define an instantaneous rate of change for any function, even if it doesn't represent the position of an object.

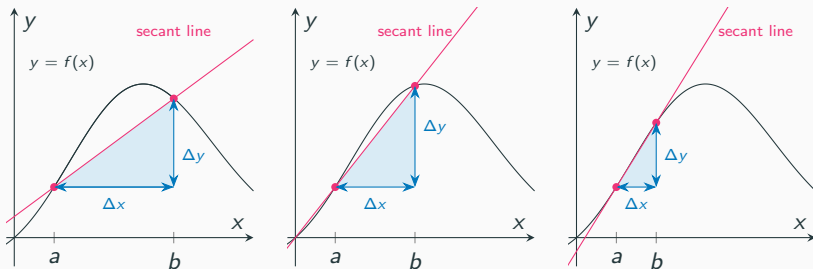
Definition

The **instantaneous rate of change** of a function f at a , is defined to be the limit of the of f over shorter and shorter intervals containing a .

We denote the instantaneous rate of change of f at a by .

Limiting secant lines geometrically

Geometrically: We can sketch a secant line through the points $(a, f(a))$ and $(b, f(b))$. As we take b closer and closer to a , the slope of the secant line will approach $f'(a)$.

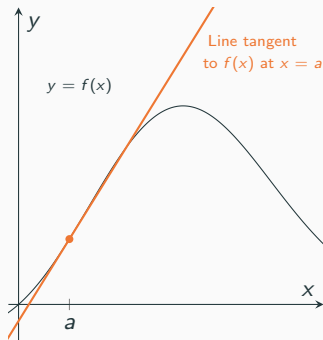


Here, $\Delta x = b - a$ and $\Delta y = f(b) - f(a)$.

Note: We could also use two points on either side of $(a, f(a))$.

Limiting secant lines geometrically

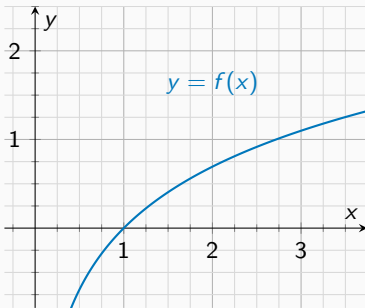
Geometrically: As the two points get closer and closer to $(a, f(a))$, the secant line looks more and more like the line **tangent** to the graph of f at $x = a$.



The instantaneous rate of change of f at a is equal to the slope of the **tangent line** at $x = a$.

Estimating the slope of a tangent line at a point

The graph of a function $y = f(x)$ is shown below.



Poll Question

Without doing any calculations, rank the following numbers from least to greatest.

$$f'(1), \quad f'(2), \quad f'(3)$$

Which one is smallest?



$f'(1)$

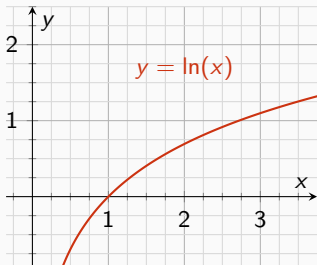


$f'(2)$



$f'(3)$

Estimating the slope of a tangent line at a point



The graph of $f(x) = \ln(x)$ is shown on the left.

<https://www.desmos.com/calculator/dwyvr6nc67>

Discussion Question

Use the graph to estimate the value of $f'(1)$. How could you make your estimate more accurate?

After Class

Approximate the values of $f'(1)$, $f'(2)$ and $f'(3)$.

Summary: Approximating Instantaneous Rates of Change

Discussion Question

For each representation of a function, what methods do we have to approximate instantaneous rates of change?

Tables

Graphs

Formulas

Words

To do before next class:

- Try questions 1-3 from the Week 2 Checkpoint Workbook.
- Please read section 2.1 carefully and do problems 1-10 from HW 2 (available on WileyPlus and due on Tuesday Sep 29 at 10pm)

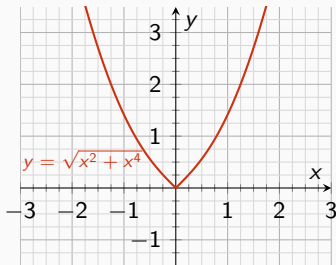
- Drop-in study hours have started and the Math Learning Centre is open.
- Got questions? The Piazza discussion board is now up and running.

Have a great day!

For the Curious

Below is the graph of $y = \sqrt{x^2 + x^4}$.

<https://www.desmos.com/calculator/ax1b8we9rj>



After Class

Use Desmos to Zoom in to the graph at the point with $x = 0$. What do you notice?